ANALYSIS REPORT

Wave 5.2R Stage 5 Complex Harmonic Coefficient Residuals Results

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Decision

Stage 5 is complete with a positive component-level result.

All `18` first-screen runs and all `4` conditional stability runs completed without a failure. `H08` is the scalar raw-error leader, but it is not the multi-index recommendation because it regresses closure, retained-amplitude, and retained-phase behavior relative to PF-A.

`H04`, the deep bounded correction to the nine PF-A core complex coefficients, passes every Stage 5 exit gate. It advances as a qualified structured component for Stage 6. It does not replace the accepted periodic GRU, does not become a new production best, and is not yet a full PINN.

Scope And Integrity

- dataset: `polished_dataset`;
- inputs: setpoints only;
- surface: `Fw`;
- accepted curves: `966`;
- split: `675` train, `194` validation, `97` test;
- angular grid: `2048` uniform points on `0 <= theta < 2*pi`;
- split signature: `c1aa8718fb9bf88cc2021c121dc4f3b4010fc1d2e45ac90af5f4376aa64f8e16`;
- runtime measured inputs: none;
- target-derived runtime inputs: none;
- official TE Curve Verification Pipeline: not run.

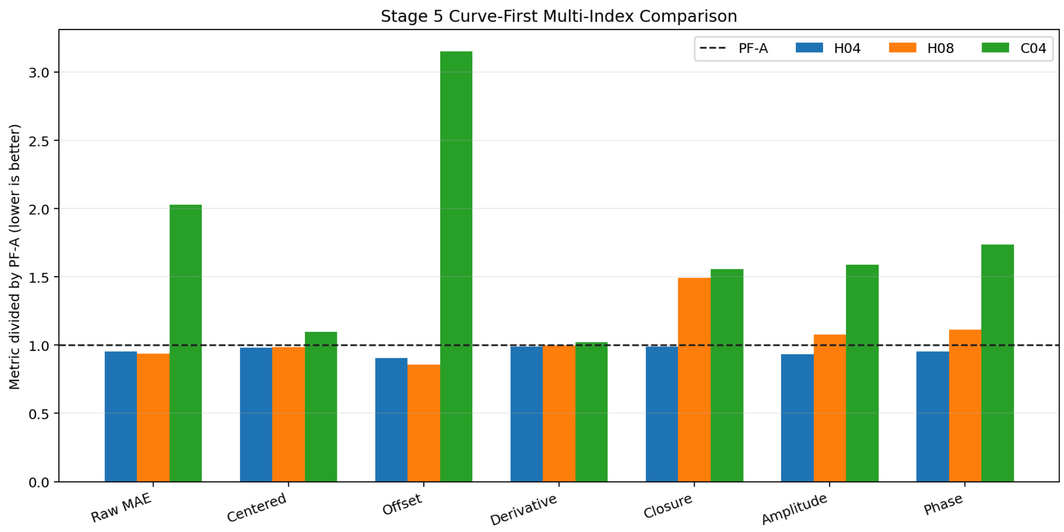
Campaign Execution

Item	Result
Planned first-screen runs	18
Completed first-screen runs	18
Failed first-screen runs	0
Conditional stability runs	4
Failed stability runs	0
Candidate formulations	5
Frozen first-screen seed	314159
Stability seeds	271828, 161803
Representation alignment	coefficient extraction, training, reconstruction, and evaluation share one uniform grid

Primary Curve-First Comparison

Metric	PF-A	C04	H04	H08	H04 vs PF-A
Raw MAE [deg]	0.001808977	0.003669246	0.001725884	0.001693343	+4.59%
Centered MAE [deg]	0.001381875	0.001518313	0.001355528	0.001363757	+1.91%
Offset error [deg]	0.000975232	0.003074926	0.000884399	0.000834846	+9.31%
Peak-to-peak error [deg]	0.004724678	0.004072875	0.004714630	0.003604733	+0.21%
Derivative MAE [deg/sample]	0.000937212	0.000957538	0.000928304	0.000936523	+0.95%
Closure error [deg]	0.000294624	0.000458545	0.000291202	0.000439377	+1.16%
Amplitude MAE [deg]	0.000194541	0.000309160	0.000181567	0.000209224	+6.67%
Phase MAE [rad]	0.337145853	0.584952853	0.321016671	0.374836453	+4.78%

H04 improves raw MAE by `4.59%` versus PF-A and by `52.96%` versus its matched direct coefficient control.



Stage 5 multi-index comparison

Why H08 Is Not The Recommendation

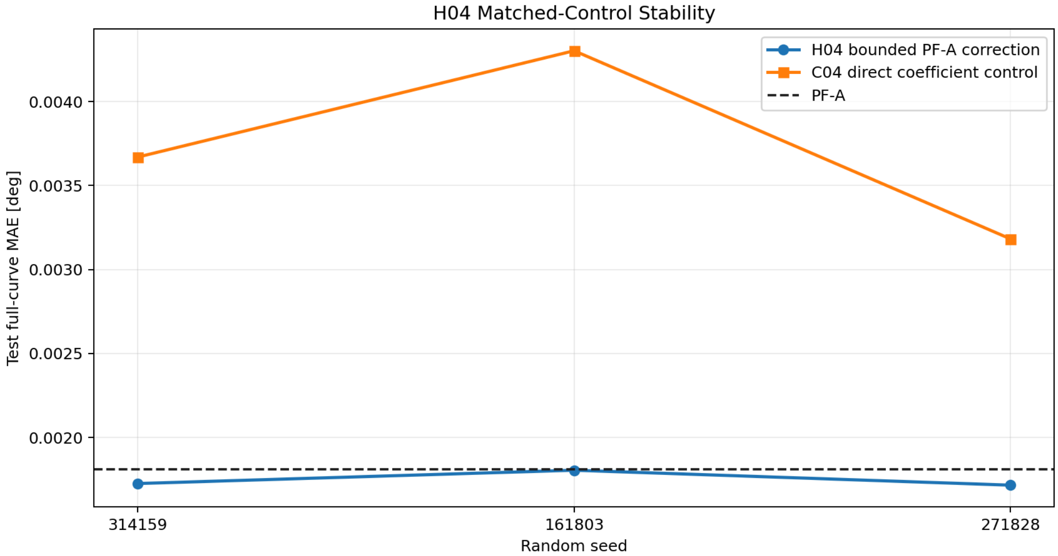
H08 reaches the lowest raw MAE, `0.001693343 deg`, but uses the broader training-selected order set. Relative to PF-A it worsens:

- periodic closure from `0.000294624` to `0.000439377 deg`;
- retained amplitude MAE from `0.000194541` to `0.000209224 deg`;
- retained phase MAE from `0.337145853` to `0.374836453 rad`. This is a real raw-error gain, not Stage 4-style analytical cancellation, but it is not the best balanced component. The later Stage 6 spectral and Sobolev work may revisit the added orders with direct derivative and spectral control.

H04 Stability

Seed	H04 MAE [deg]	C04 MAE [deg]	H04 vs PF-A
314159	0.001725884	0.003669246	+4.59%
161803	0.001805115	0.004302256	+0.21%
271828	0.001716240	0.003182218	+5.13%

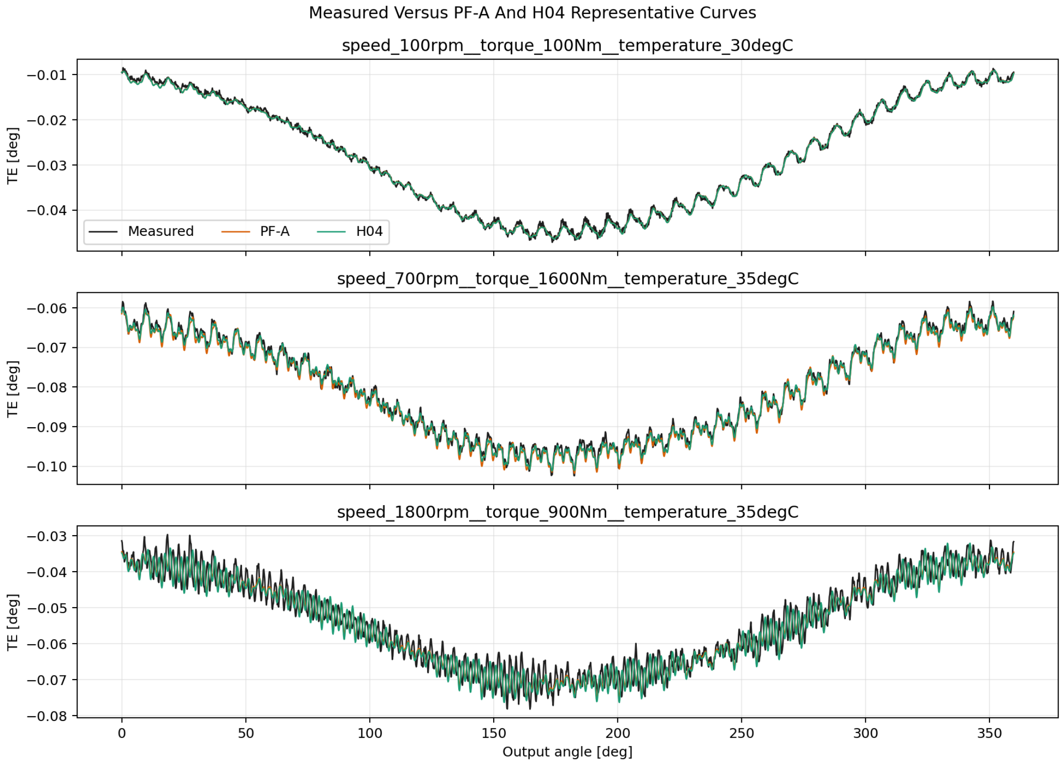
H04 mean MAE across the three seeds is `0.001749080 deg`, with standard deviation `0.000039818 deg`. Every seed beats PF-A and its matched direct C04 control.



H04 stability

Representative Full Curves

The following deterministic test cells compare measured TE, frozen PF-A, and the first-screen H04 checkpoint on the same `2048` -point grid.



Representative H04 curves

Exit Gates

All ten gates listed below pass.

Gate	Candidate	Reference or limit	Result
Raw MAE vs PF-A	0.001725884	0.001808977	passed
Raw MAE vs C04	0.001725884	0.003669246	passed
Centered MAE vs PF-A	0.001355528	0.001381875	passed
Offset vs PF-A	0.000884399	0.000975232	passed
Derivative vs PF-A	0.000928304	0.000937212	passed
Closure vs PF-A	0.000291202	0.000294624	passed
Amplitude vs PF-A	0.000181567	0.000194541	passed
Phase vs PF-A	0.321016671	0.337145853	passed
Correction energy	0.006915951	0.500000000	passed
Three-seed worst vs PF-A	0.001805115	0.001808977	passed

Scientific Interpretation

Stage 5 demonstrates that physics-informed assistance can work in this dataset when the analytical and learned parts share the same representation and the learned freedom is constrained to explicit complex coefficients.

The result is stronger than a generic Fourier feature observation:

- the PF-A analytical contribution remains explicit;
- the network learns only bounded coefficient corrections;
- zero correction replays PF-A exactly;
- corrections remain below one percent of anchor RMS for H04;
- every retained harmonic contribution is inspectable;
- the gain survives three seeds and matched direct controls.

  The result still does not prove a differential-equation PINN. H04 is a qualified grey-box structured component that Stage 6 can augment with spectral and Sobolev guidance.

Program Decision

- Stage 5 status: complete, positive component-level result;
- qualified component: H04 bounded PF-A core-coefficient correction;
- raw-error-only leader: H08;
- production/model-registry promotion: no;
- accepted periodic GRU replacement: no;
- stability continuation: complete;
- official TE Curve Verification Pipeline: deferred from normal closeout;
- next executable step: Stage 6, Spectral And Sobolev Guidance.

Artifact Map

- campaign: `output/training_campaigns/2026-07-28-16-17-06_wave52r_stage5_complex_harmonic_coefficient_residuals_2026_07_28/` ;
- representation and exit gates: `output/analysis/wave_5_2r/stage5_complex_harmonic_coefficient_residuals/` ;
- H04 first-screen run: `output/training_runs/complex_harmonic_coefficient_residuals/2026-07-28-16-17-13__stage5_h04/` ;
- [Stage 5 model report] (../../../analysis/model_development_waves/wave_5_2/physics_guided_pinn_reassessment/%5B2026-07-28%5D/stage5_complex_harmonic_coefficient_residuals/stage5_complex_harmonic_coefficient_residual_model_report.m
- preliminary-screen integrity: the superseded sixteen-run screen was archived recoverably under `.temp/stage5_superseded_16_run_screen/` after the missing data-selected direct controls were detected; only the corrected eighteen-run campaign is canonical.